When did plants first live on land? What the rocks can and cannot tell us

Barry W. Brook (University of Tasmania), with Claude Opus 5.5 (Anthropic) and OpenAI Codex. Draft of 5 November 2026; a plain-language summary of a living research repository, https://github.com/BWBrook/silurian_plants.

The puzzle

Every forest, grassland and crop descends from a few small plants that left the water hundreds of millions of years ago. Land plants changed the air, the soils and the climate, and built the stage on which animals came ashore. Yet scientists disagree about when this happened by tens of millions of years.

One way to date it is a *molecular clock*: compare the DNA of living plants and work out how long their lineages have been diverging. Clocks need fossils to set the time scale, and they lean heavily on one kind of fossil argument: a *maximum*. A maximum says "if this group had existed before a certain date, we would have found its fossils in older rocks, and we have not". Change the maximum and the clock's answer changes with it.

Three recent clock studies show how much rides on this. Two place the common ancestor of all living land plants 515–474 million years ago (Ma), in the Cambrian. The newest, using a different set of fossil maxima, places it at 457–450 Ma (Morris et al. 2018; Harris et al. 2022; Hooper et al. 2026). For the ancestor of living *vascular* plants, the group with internal plumbing that includes ferns, conifers and flowering plants, estimates run from about 452 to 428 Ma (figure below).

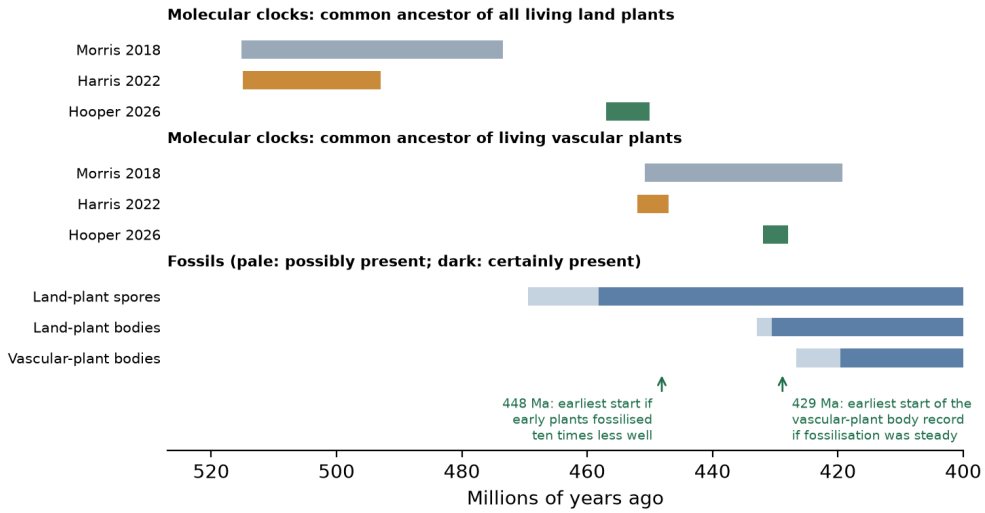


Figure. Molecular-clock estimates for two key common ancestors, set against what the rocks show. Green arrows: how far back the record of vascular-plant bodies could reach under two assumptions about early fossilisation; the fossils alone cannot choose between them.

Why the rocks are hard to read

The oldest traces of land plants are spores: tough, microscopic, and made in huge numbers. They appear in rocks around 460 Ma. Fossils of the plants themselves (stems and spore capsules, often only millimetres long) appear about 30 million years later. Early land plants were small, lacked wood, grew in places that rarely became rock, and rotted easily. So the absence of plant fossils from older rocks might mean the plants were not there, or only that they were not preserved and found. A maximum has to assume the first.

What we did

We asked a simple question: for each of the three maxima that divide the clocks (515.5, 459.1 and 431.2 Ma), what can the actual fossil record test? We used three bodies of evidence.

- **Spores from Libyan boreholes.** In one North African basin, 28 rock samples older than about 443 Ma contain no spores of the advanced (trilete) type, while 51 of 59 younger samples do. This is the only place we found where a "not found in older rock" claim can be tested with numbers.

- **Spores from a Saudi Arabian borehole**, as a rougher check.
- **The dated fossils of early plant bodies** that we could link to a place and an age: 77 dated places, 50 of them with members of the vascular-plant group. From how these finds are spread through time, a standard statistical method estimates how far back before the oldest find the fossil record of a group probably began. That estimate depends on how well plants were fossilised before the oldest find, so we tried several assumptions and reported each one.

What we found

1. **The oldest maximum (515.5 Ma) cannot be tested from the rocks we could assess.** It rests on whether some Cambrian microfossils are land-plant spores, and the specialists who described them do not accept them as such. Even if they were, they are younger than the maximum.
2. **The 459.1 Ma maximum gets one real test, and it is conditional.** In that one Libyan basin, the samples rule out producers of advanced spores that were abundant throughout the time before 443 Ma and whose spores reached the rocks as well as later ones did. They cannot rule out rare producers, or spores that simply did not reach and survive in those rocks.
3. **For vascular plants, the answer depends most on one assumption.** If plants were fossilised at a steady rate, their body fossils became findable no earlier than about 429 Ma, which agrees with the youngest clock. In one illustrative case where early plants fossilised ten times less well before about 427 Ma, which is plausible but unproven, the same finds allow a start as early as 448 Ma, which agrees with the older clocks. The fossils we have cannot tell these apart, because the answer is decided in a stretch of time that left almost no plant bodies. Fossil ages, which finds count as vascular plants and whether finds are independent matter too.
4. **One fossil can move the answer by 20 million years.** Land-plant spores are certainly present by 458 Ma. Counting stems and spore capsules, the same method says land-plant bodies became findable 21–24 million years later. Count one set of tiny, spore-filled capsule fragments from Late Ordovician Oman as well, and the gap almost vanishes. Plants may have lived for a long time before their bodies show up as fossils, but the record cannot say how long.

What it means

An older, sparsely fossilised early flora is *allowed* by the evidence, but not *required* by it. The young clock fits the record under the simplest assumptions; the older clocks fit only if early plants were rare or poorly preserved for a long time, or if their recognisable features came well after their origin. Neither can be excluded today. The practical lesson for clock studies is to treat these fossil maxima as conditional on stated assumptions about preservation and plant anatomy, and to carry that uncertainty into the final dates instead of hiding it.

How this work was done

The study was carried out by two AI systems, Claude (Anthropic) and Codex (OpenAI), working as research collaborators with Barry Brook, who set its direction. The AI systems read and checked the sources, built and tested the models, reviewed each other's work and drafted the text. Every number traces to a file and script in the public repository, which is the primary record of the work; this summary, the full paper and any journal version are snapshots of it.

Sources

Harris, Brogan J., James W. Clark, Dominik Schrempf, Gergely J. Szöllősi, Philip C. J. Donoghue, Alistair M. Hetherington, and Tom A. Williams. 2022. "Divergent Evolutionary Trajectories of Bryophytes and Tracheophytes from a Complex Common Ancestor of Land Plants." *Nature Ecology & Evolution* 6 (11): 1634–43. https://doi.org/10.1038/s41559-022-01885-x.

Hooper, Jack, Annabel M. Worth, Charles H. Wellman, James W. Clark, and Philip C. J. Donoghue. 2026. "Reconstructing Spore Evolution Recalibrates the Timescale of Land Plant Diversification." *Nature Plants*. https://doi.org/10.1038/s41477-026-02432-3.

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